

Solution Requirements

Date	15 March 2026
Team ID	SWTID-2026-9392
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	User registration and secure login via Firebase/OAuth (Google Sign-In) to maintain user workspaces and saved comic projects.	High
2	Authorization levels	Role-based access control: User (create, edit, and export personal comics) and Admin (monitor platform usage, manage API usage limits, and view feedback).	Medium
3	External interfaces	Integration with Google Gemini API (for story plot, script, dialogue, and image generation prompts) and cloud storage (AWS S3 / Firebase Storage) for image hosting.	High
4	Transactions processing	Asynchronous task processing for sequential panel image generation and dynamic text rendering on comic layouts.	High
5	Reporting	Personal dashboard displaying user history, generated stories, saved drafts, and exported comic formats (PDF/ PNG).	Medium
6	Business rules, etc.	Maximum panel generation limit per project (e.g., up to 6 panels per story page); strict input prompt filtering for safety and content policy adherence.	High
7	Compliance to laws or regulations	Adherence to standard data protection regulations (GDPR/CCPA) for user profile data and copyright compliance regarding AI-generated art assets.	Medium
8	Other		Low

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
		Ability to re-generate individual panel images or edit auto-generated dialogues before final export.	

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Rapid response time for script generation and smooth layout rendering.	Page loads / Script generated in < 3 seconds; panel rendering < 5s.
2	Scalability	System infrastructure built to handle horizontal scaling for concurrent user story generation requests.	Supports up to 10,000 concurrent active users.
3	Security & Data Privacy	Encrypted API keys, tokenized authentication sessions, and secure HTTPS protocol for all payload delivery.	End-to-end AES-256 encryption & TLS 1.3 protocol.
4	Reliability & Availability	High uptime and automatic recovery mechanisms for third-party API rate limiting or timeouts.	99.9% uptime per month with graceful fallback handling.
5	Usability & Accessibility	Responsive web user interface tailored for both desktop and mobile screen viewports with clear comic layout tools.	Complies with WCAG 2.1 AA standards.
6	Other	Maintainability, portability across standard web browsers without requiring external browser extensions.	High Maintainability, Portability across all modern browsers.